

Airfoil CFD Analysis using ANSYS Fluent

Project Overview

This project presents a Computational Fluid Dynamics (CFD) simulation of airflow over a 2D airfoil using ANSYS Fluent.

The objective was to study airflow behavior around the airfoil and observe the pressure and velocity distribution generated during the simulation.

Objective

The main goals of this project were:

- Understanding the basics of CFD workflow.
 - Learning how to create and prepare geometry for simulation.
 - Generating a suitable mesh for airflow analysis.
 - Applying boundary conditions inside ANSYS Fluent.
 - Observing airflow behavior around an airfoil.
 - Analyzing velocity and pressure contours.
 - Understanding the relation between air velocity and pressure distribution over the wing section.
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Software Used

- ANSYS Fluent
 - ANSYS Workbench
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Simulation Workflow

1. Geometry Preparation

A 2D airfoil geometry was prepared inside ANSYS Workbench to represent a simplified wing section for aerodynamic analysis.

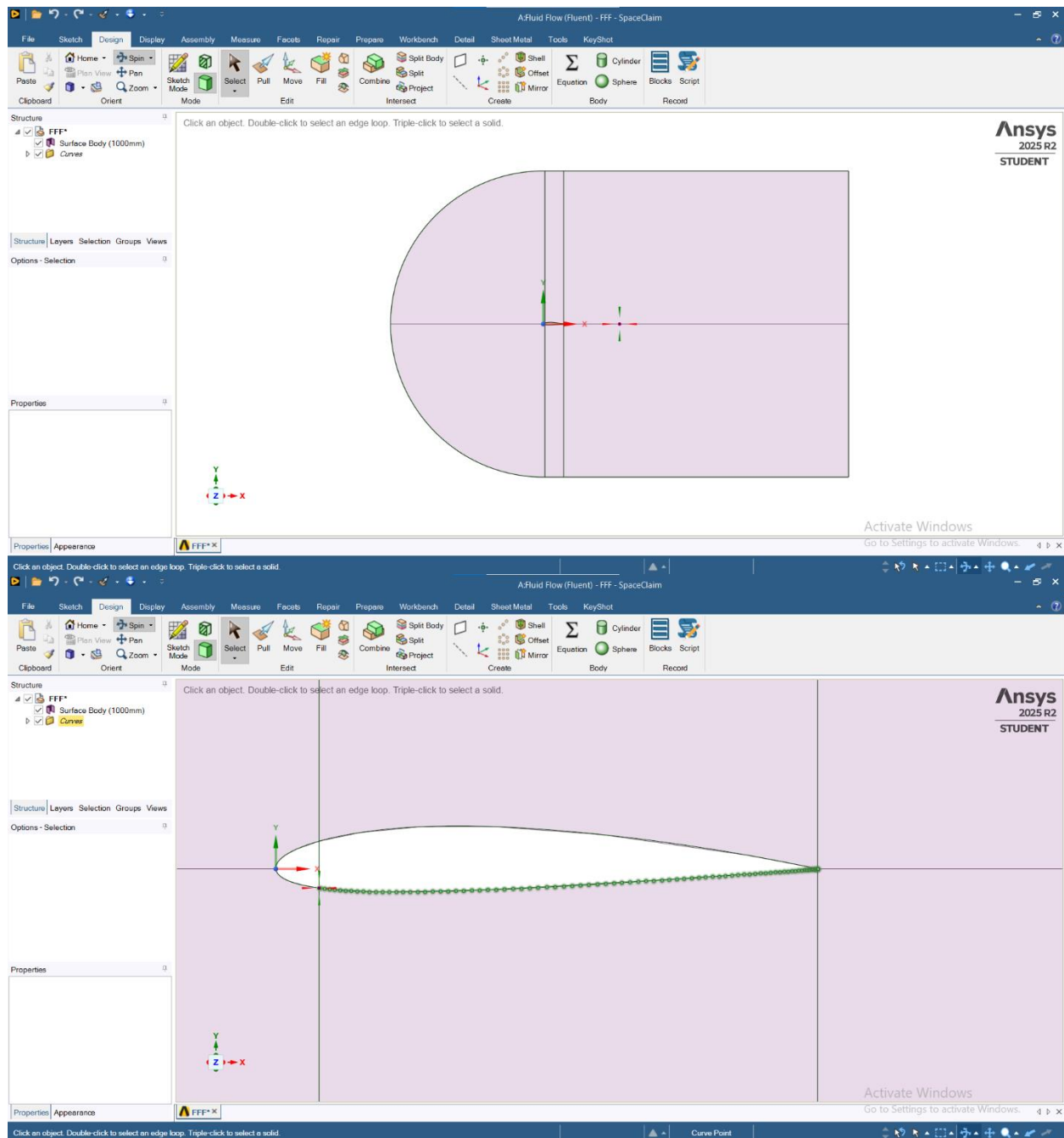


Figure 1: airfoil geometry

2. Mesh Generation

A computational mesh was generated around the airfoil to divide the flow domain into smaller elements suitable for numerical analysis.

Special attention was given to:

- Mesh distribution near the airfoil surface.
- Refinement around expected high-gradient regions.
- Overall mesh quality.

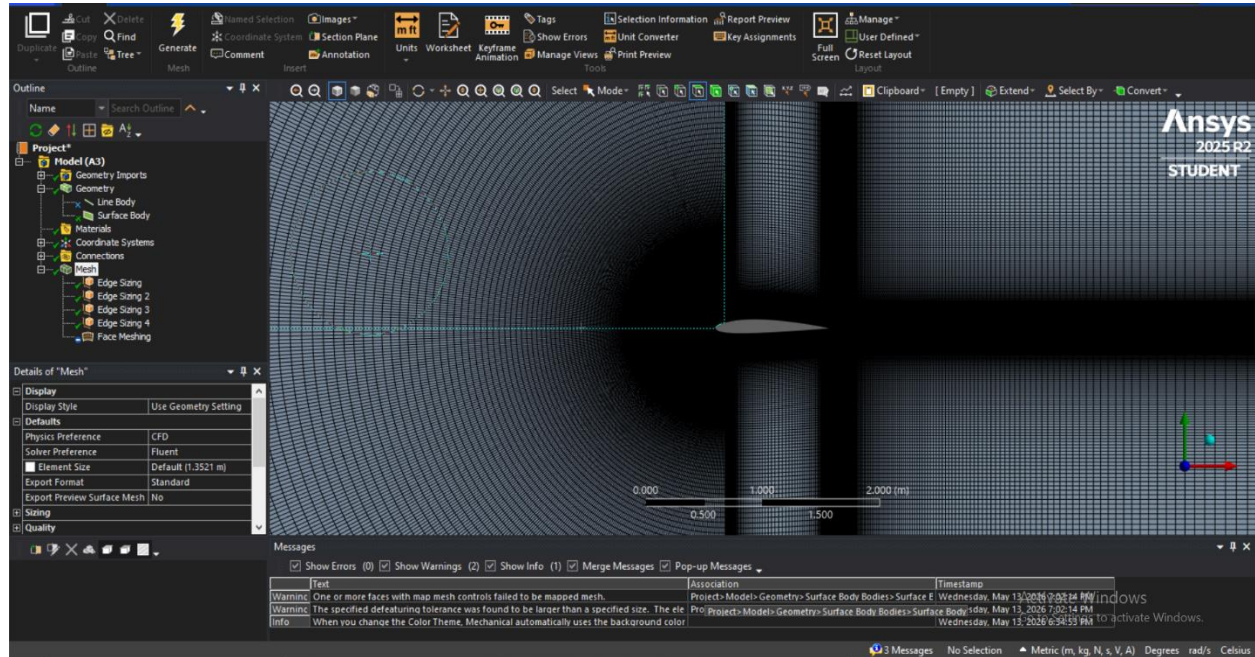


Figure 2: mesh distribution

Boundary Conditions

The following boundary conditions were applied:

- Velocity Inlet
- Pressure Outlet
- Air as working fluid
- Steady-state flow conditions

These settings allowed simulation of airflow passing around the airfoil section.

 Velocity Inlet ×

Zone Name

Momentum

Thermal

Radiation

Species

DPM

Multiphase

Potential

Structure

UDS

Velocity Specification Method

Components

Reference Frame

Absolute

Supersonic/Initial Gauge Pressure [Pa]

0

X-Velocity [m/s]

45.6

Y-Velocity [m/s]

0

Turbulence

Specification Method

Intensity and Viscosity Ratio

Turbulent Intensity [%]

5

Turbulent Viscosity Ratio

10

Apply

Close

Help

 Pressure Outlet ×

Zone Name

Momentum

Thermal

Radiation

Species

DPM

Multiphase

Potential

Structure

UDS

Backflow Reference Frame

Absolute

Gauge Pressure [Pa]

0

Pressure Profile Multiplier

1

Backflow Direction Specification Method

Normal to Boundary

Backflow Pressure Specification

Total Pressure

☐ Prevent Reverse Flow

☐ Average Pressure Specification

☐ Target Mass Flow Rate

Turbulence

Specification Method

Intensity and Viscosity Ratio

Backflow Turbulent Intensity [%]

5

Backflow Turbulent Viscosity Ratio

10

Apply

Close

Help

Figure 3: boundary conditions

Solver Setup

The simulation was performed using:

- Pressure-based solver
- Steady-state analysis
- Standard turbulence model
- Default air properties

Iterations were executed until convergence behavior became stable.

Results and Discussion

Velocity Contours

The velocity contour shows airflow acceleration around the airfoil surface.

Higher velocity regions appeared over parts of the airfoil due to aerodynamic effects.

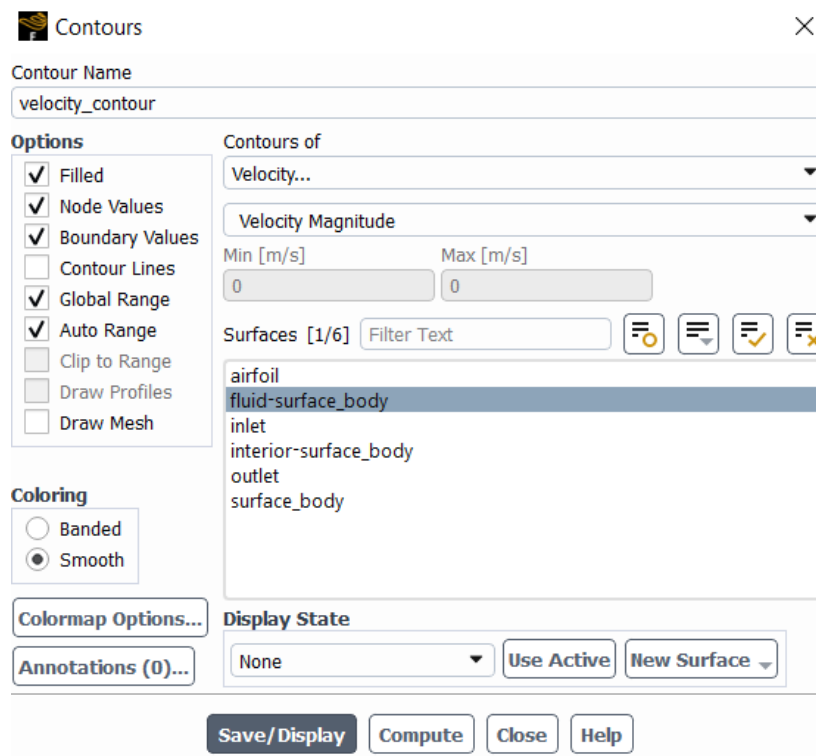


Figure 4: velocity contour

Pressure Distribution

Pressure contours demonstrated pressure variation around the airfoil.

Pressure differences between upper and lower surfaces contribute to lift generation.

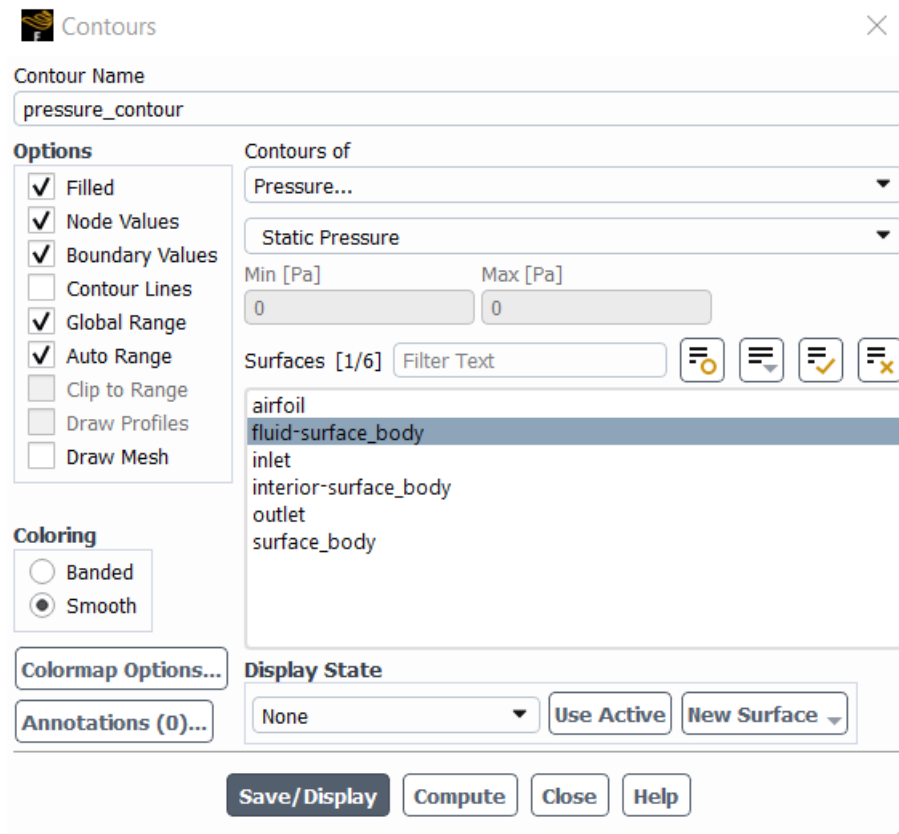


Figure 5: pressure contour

Flow Behavior Observation

The simulation helped visualize:

- Streamline behavior
- Velocity changes
- Pressure variation around the wing profile

This provided a practical introduction to aerodynamic CFD analysis.

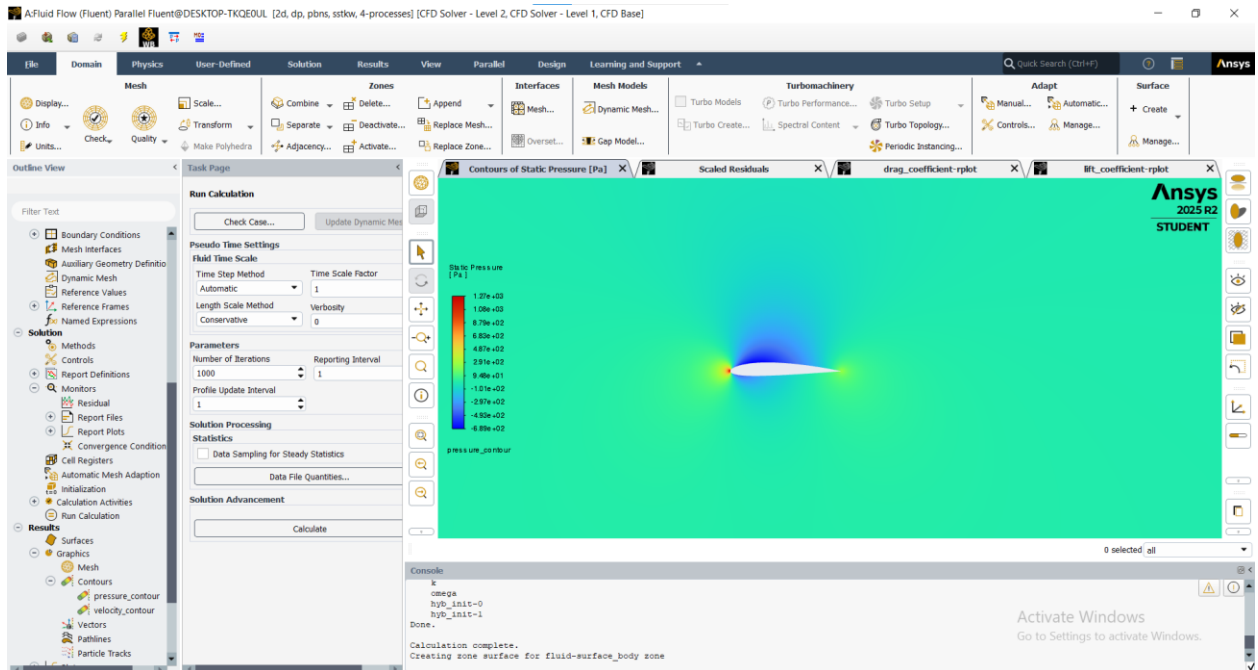


Figure 6: pressure variation

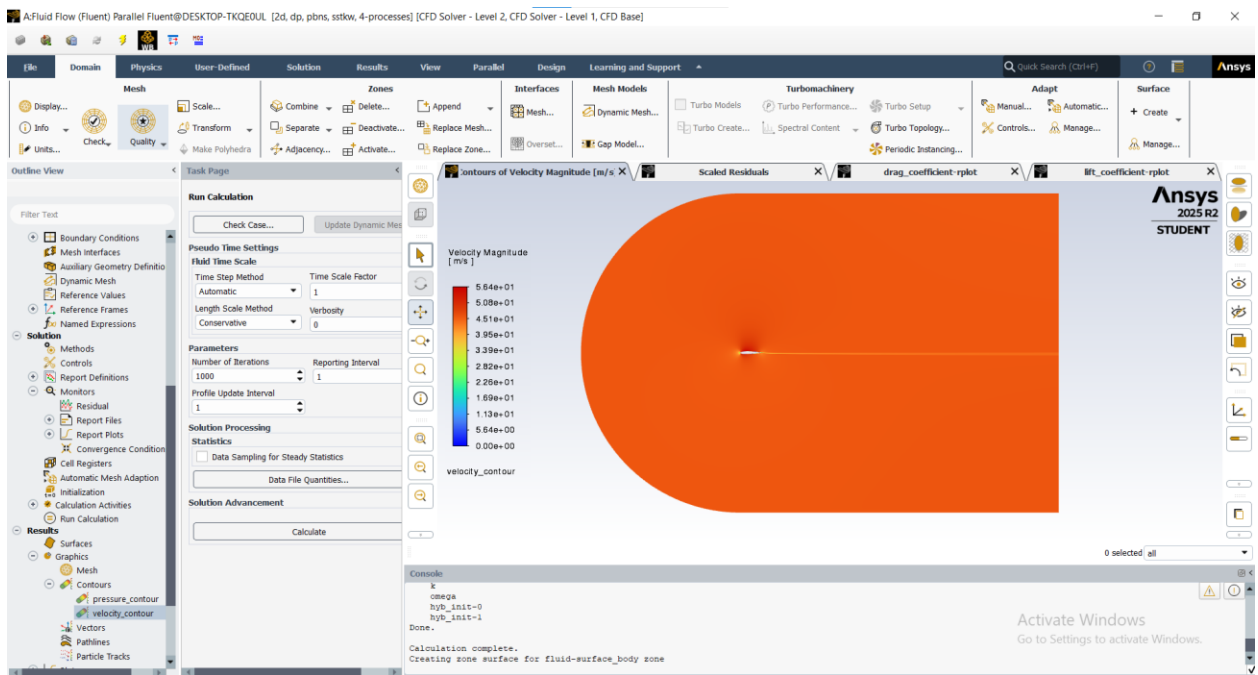


Figure 7: velocity variation

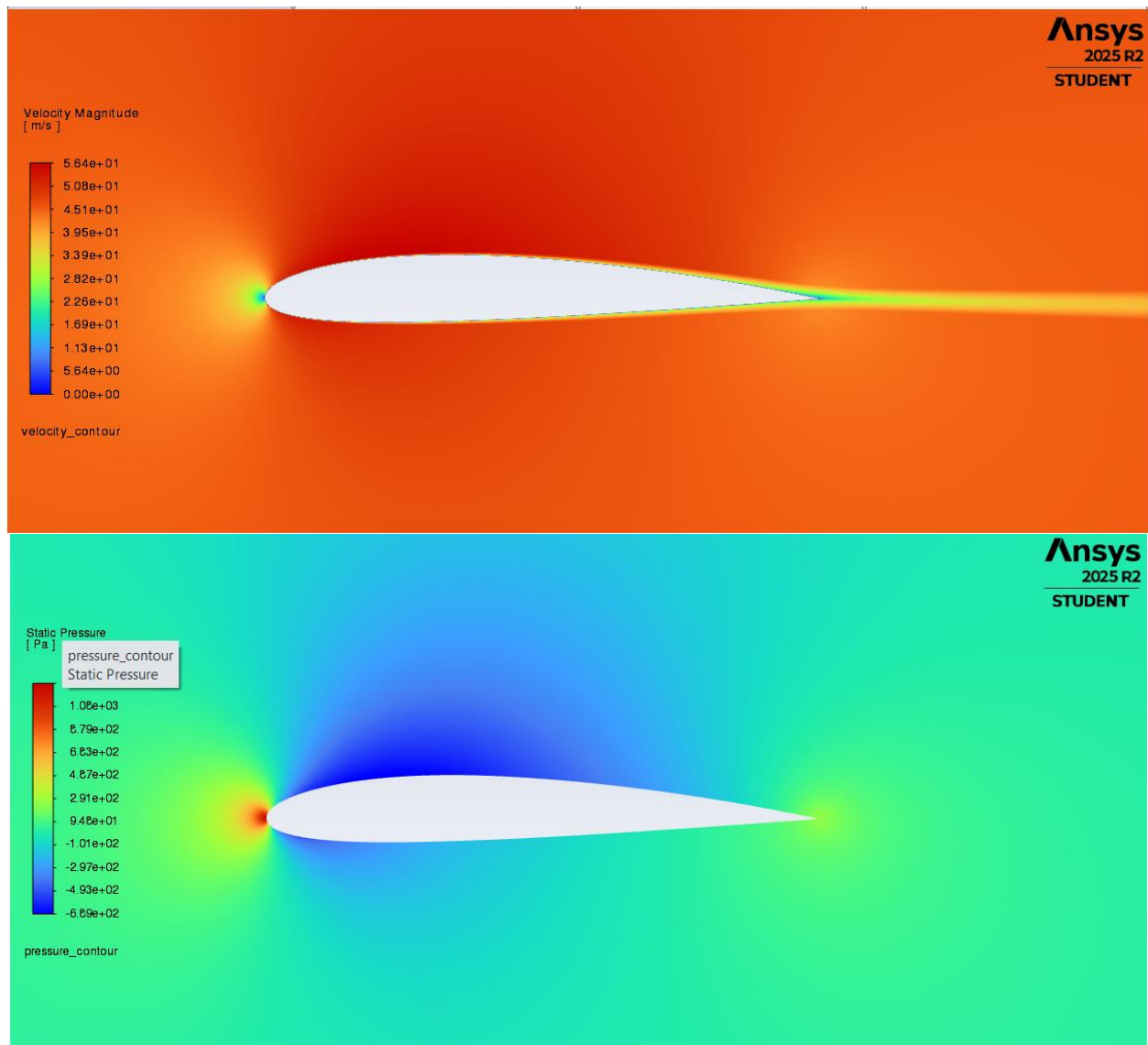


Figure 8: velocity and pressure variation in closer

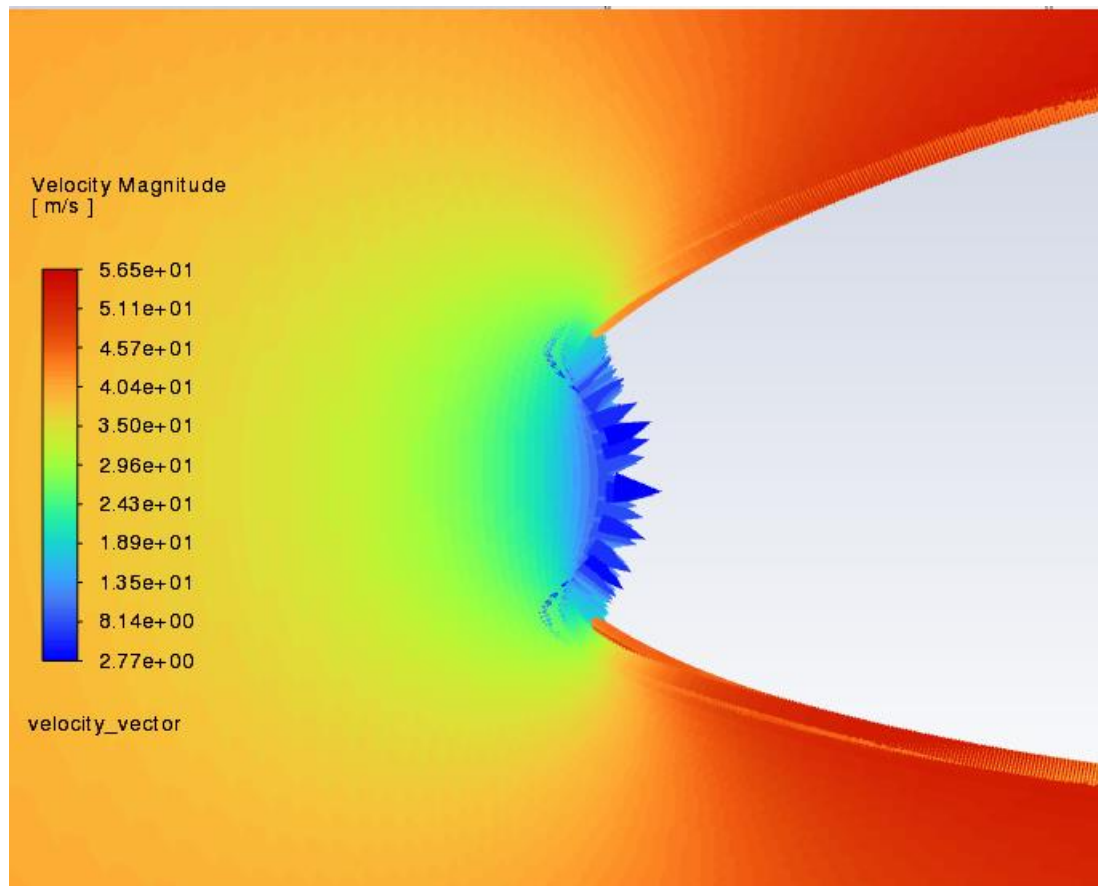


Figure 9: stagnation effect

Challenges Faced

During the project, several challenges were encountered, including:

- Understanding CFD workflow steps.
- Adjusting mesh settings.
- Selecting suitable boundary conditions.
- Interpreting simulation results correctly.

These challenges improved practical understanding of simulation procedures.

What I Learned

Through this project, I learned:

- Fundamentals of Computational Fluid Dynamics (CFD).
 - Basic aerodynamic analysis concepts.
 - Geometry and mesh preparation techniques.
 - Applying boundary conditions in ANSYS Fluent.
 - Reading velocity and pressure contours.
 - Understanding airflow behavior around airfoils.
 - Importance of mesh quality in simulation accuracy.
 - Basics of engineering simulation reporting and result presentation.
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Conclusion

This project provided an introductory practical experience in aerodynamic CFD simulation using ANSYS Fluent.

It helped build foundational knowledge in airflow analysis and engineering simulation workflow, forming a strong base for more advanced future projects in fluid mechanics and aerospace-related applications.

Future Improvements

Possible future developments for this project include:

- Testing different airfoil shapes.
- Comparing multiple angles of attack.
- Performing transient simulations.
- Studying lift and drag coefficients quantitatively.
- Using finer mesh refinement for higher accuracy.